

Shark Tank Pitch Analyzer - Rephrased Report

1. Problem Statement

Entrepreneurs frequently find it challenging to assess the quality of their pitch delivery, particularly regarding vocal tone, confidence, clarity, pacing, and persuasion style. Current evaluation tools primarily focus on textual analysis, neglecting critical verbal elements. Additionally, founders often lack objective insights into whether their pitch aligns with key business criteria such as market understanding, revenue model robustness, competitive positioning, and overall feasibility.

The objective of this project is to develop a multimodal AI framework that evaluates both the audio delivery and business content of a pitch, providing tailored feedback through virtual investor personas to simulate a Shark Tank-like assessment.

2. System Overview

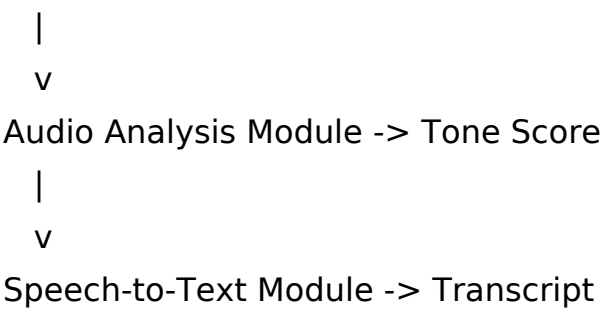
The SharkTank_Pitch_Analyzer is structured around three core modules:

- Voice & Tone Analysis: Examines audio attributes to measure clarity, energy, confidence, pauses, filler word usage, and pitch modulation.
- Content & Business Logic Analysis: Utilizes ASR for transcription and NLP-based heuristics to assess problem clarity, solution explanation, market opportunity, revenue model, and competitive awareness.
- Virtual Shark Panel Feedback: Emulates a panel of investor personas offering feedback based on tone and business evaluations. Personas include the Visionary, Finance Shark, Customer Advocate, and Skeptic.

The system delivers a comprehensive report combining tone scores, business scores, persona feedback, and an overall recommendation.

3. Pipeline Architecture

Audio Input (.wav or .mp3)



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v
NLP Content Analysis -> Business Score

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Virtual Shark Panel -> Persona Feedback

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v
Combined Analysis Report -> Final Recommendation

4. Voice & Tone Analysis

Audio Features Extracted: Average pitch and variation, pause frequency, energy fluctuations, filler word count, and duration.

Tone Scoring: Heuristic scoring (0-100) based on clarity, confidence, energy, and filler word use.

Example Output:

Tone Score: 71 / 100

Strengths: Strong volume and pacing, effective pitch modulation

Weaknesses: Minor use of filler words, could enhance expressiveness

5. Business Content Evaluation

Transcription: Conducted using Whisper ASR or a fallback method.

Content Scoring: Evaluates problem clarity, solution articulation, revenue indicators, market opportunity, and competitive awareness.

Pitch Structure Detection: Hook -> Problem -> Solution -> Ask.

Business Score: 0-100

Example Output:

Business Viability Score: 76 / 100

Detected Structure: Hook, Problem, Solution, Ask

Revenue Signals: 2, Market Signals: 2

6. Virtual Shark Panel Feedback

Persona Feedback Examples:

- Visionary: "Promising concept with scalable potential. Consider including growth projections."
- Finance Shark: "Revenue indicators are visible. Clarify unit economics, CAC, and

margins."

- Customer Advocate: "Problem is evident. Include customer examples or pilot data for stronger credibility."
- Skeptic: "Execution and competitive risks are unclear. Minimize filler words for more confident delivery."

Final Recommendation: INVEST (conditional)

7. Sample Input and Output

Input Audio: sample_data/pitch1.mp3 (or .wav)

Output Report: demo_outputs/pitch1_analysis_report.txt (representative demo output provided)

8. Limitations and Future Work

- Current system relies on lightweight heuristics; incorporating deep learning could enhance scoring accuracy.
- ASR transcription accuracy may suffer in noisy conditions; fine-tuned models could improve performance.
- Persona feedback can be enriched using LLM-based generation for more realistic investor commentary.

9. Conclusion

SharkTank_Pitch_Analyzer offers a comprehensive multimodal evaluation of startup pitches by combining voice tone assessment, business content analysis, and persona-based feedback. It equips entrepreneurs with actionable insights, closely resembling a Shark Tank panel evaluation, enabling effective pitch practice.

Report prepared as part of the SharkTank_Pitch_Analyzer project submission.